

EDUCATION

• **Booth School of Business, University of Chicago, IL** **September 2026-Present**

PhD in Economics

• **Michigan State University, East Lansing, MI** **GRE: 170Q/166V** **May 2024**

Honors Dual Degree in Economics and Molecular Genetics & Genomics, Minors in Mathematics and Law & Public Policy

•Additional Academic Achievements

NSF Graduate Research Fellow: for scholars with demonstrated potential to make significant contributions to research

Doctoral Centennial Fellow: for exceptional promise as a future scholars, awarded by the Booth Economics Department

George F. McGregor Scholar: for econometric ability, awarded by Professor Wooldridge and Professor Soren Anderson

First Place UURAF Poster Presentation: for Sexual Dimorphism in Lupus

MSU Economics PURI 2022 Grant Winner: provided top research proposals with 1 year of funding

MSU Econ Scholars Program: recognizing top economics undergraduates

Dean's List every semester in the college of Natural Science and the college of Social Science

Hanna M. Rasmussen Memorial Alumni Distinguished Scholar: full-ride merit-based scholarship to Michigan State University

RESEARCH EXPERIENCE

• **Stores Economics and Applied Science (SEAS) at Amazon** *June 2024-Present*

Economist Apprentice

Amazon, Seattle, WA

– Research with Dr. Nichols, Dr. Jahedi, Professor Imbens, and Professor Card on network externalities, labor productivity, worker health, consumer spending, and high-dimensional causal inference methods.

• **Liang Laboratory** *September 2022-2025*

Professorial Assistantship

Michigan State University, East Lansing, MI

– Second author for a paper on sex-linked lupus bias in XIST and RPII KD fetal mouse keratinocytes–2025 publication

• **MSU Department of Economics** *September 2022-2023*

Provost's Undergraduate Research Initiative Researcher

Michigan State University, East Lansing, MI

– Research on incorporation of Kahneman and Tversky prospect theory for a model of symmetric information with a sender-specific signal for a receiver taking a non-contractible action affecting the welfare of both players.

• **MSU Department of Mathematics** *September 2022-2023*

Honors Research Project

Michigan State University, East Lansing, MI

– Honors Project on Graph Theory and properties of Cayley graphs in Abstract Algebra and Number Theory

• **Robison Laboratory** *September 2021-2022*

Professorial Assistantship

Michigan State University, East Lansing, MI

– Received an NSP grant for analysis of hypothalamic cell deviation, drug dependency, cPTSD, and reward-seeking behavior

• **Pyeon Laboratory** *September 2020-2021*

Professorial Assistantship

Michigan State University, East Lansing, MI

– Protein assays and genetic analysis around chemokines associated with HIV and cervical cancer in human cell lines.

• **Nobrega Laboratory** *June-August 2019*

Research Internship

Human Genetics Department, University of Chicago, Chicago, IL

– Research with Dr. Nobrega examining the role of epigenetics on obesity, analyzing RNA sequencing data using R.

WORK EXPERIENCE

• **MSU Department of Economics** *September 2022-May 2024*

Undergraduate Social Media Coordinator

East Lansing, MI

– Management of the Department's social media pages—posting relevant economics literature/departamental content

• **MSU College of Social Science** *November 2023-May 2024*

Pathway to Research Program Student Mentor

East Lansing, MI

– Mentor for first-generation and diverse social science students—helping them engage in research, find fields and mentors in which they're interested, communicate with potential mentors, and secure a research position.

• **Oliver Wyman** *July 2023-August 2023*

Core Consulting Group Intern

New York City, NY

– Created spend models, slide decks, and market evaluations of metals for an aerospace engineering merger.

– Tutor/mentor for students entering consulting, focusing on networking for diverse talent with MSU SCA

• **Johnsen Fretty and Co** *May 2023-July 2023*

Mergers and Acquisitions Intern

Stamford, CT

– Evaluated value and purchasing models of potential acquisitions, particularly in adjustments for COVID impact.

• **MSU Department of Computational Mathematics, Science, and Engineering** *December 2021-May 2022*

Undergraduate Teaching Assistantship

East Lansing, MI

– Assistantship geared toward undergraduate tutoring of introductory theoretical statistics and probability.

TECHNICAL SKILLS AND INTERESTS

Languages: Fluent in Hindi, Marwari, and Gujarati and proficient in Latin and Spanish

Coding Languages: Skilled in Python, R, SQL, Spark, and STATA

Markup/Document/Replication Tools: Skilled in LaTeX, markdown, Jupyter NB, VSCode, Git, & Github

Coursework: Behavioral Economics, Advanced Econometrics, Differential Equations, Linear Algebra, Abstract Algebra, Real Analysis(Harvard Extension School), Economics of Labor & Markets, Theory of Probability & Theory of Statistics

POSITIONS OF LEADERSHIP

•**Undergraduate Liaison**, MSU Economics Department's DEI Initiative Committee *January 2022-May 2024*

•**Vice President**, Women and Minorities in Economics (WAMIE) *January 2022-May 2024*

RESEARCH PAPERS/PRESENTATIONS

The Impact of Abortion Bans After Dobbs v. Jackson on Teen Birthrates and Education Attainment: Working Paper
We find statistically significant evidence that abortion bans increase teen birth rates in the United States, using Wooldridge's new ETWFE specification to address heterogeneous pre-trends. However, concurrent COVID-19 policies complicate the analysis of abortion bans' effects on educational attainment, as states with bans also implemented less stringent pandemic restrictions, requiring further research to isolate causal effects.

Bayesian Persuasion and More: Applying Prospect Theory to Information Signaling Games: Symposium poster
This research investigates how incorporating Kahneman and Tversky's prospect theory into the Kamenica and Gentzkow Bayesian Persuasion model affects sender-receiver signaling games, hypothesizing that probability distortion breaks down the original model's predictions in single-play scenarios while maintaining similar aggregate results. Through analysis of probability distortion and diminishing sensitivity in a prosecutor-judge framework, we demonstrate that prospect theory provides a theoretical explanation for why Bayesian Persuasion rarely occurs in real-world environments, as distorted probability perceptions can fundamentally alter receiver decision-making.

TNF α Exposure in XIST and RPII KD Lupus-Prone Mice Stimulates Sex-Bias: Manuscript to be published in 2025
We investigate the role of XIST RNA in inflammation and lupus pathogenesis, as sexual dimorphism in immune regulation results in women suffering 8 out of 9 lupus cases. Through qPCR analysis and immunohistochemistry of female mouse keratinocytes with XIST knockdown treated with TNF- α inflammatory factor, we identified significant differential expression of inflammation-related genes (including NF κ BIA, TNFSF11, and NF- κ B pathway components), preliminarily mapping an epigenetic relationship between XIST RNA and altered inflammatory protein expression that may explain sex-based differences in autoimmune disease incidence.

Nesterov's Implicit Regularization Under Stochastic Gradient Descent: Working Paper
We study Nesterov's Accelerated Gradient (NAG), finding that it exhibits stronger implicit regularization than standard Gradient Descent and Polyak's Momentum, with the regularization strength increasing by factors of $((\frac{1+\beta}{1-\beta})^2)$ and $(1 + \beta)$ respectively. Through theoretical analysis using existing ODE methods to calculate error bounds and experimental validation, we show that higher momentum parameters in NAG reduce variation and enhance test accuracy by inducing a differentiable trajectory with a more powerful implicit regularizer.

Skippy FC-FC Cost Curve Live Experiment Evaluation: Manuscript submitted to the Amazon Machine Learning Conference (AMLC)
Amazon Confidential.

Amazon City: Best Economics Paper at Amazon-internal Consumer Science Summit (Last Author)
We conduct a four-month field experiment eliminating the \$25 minimum purchase requirement for subsame day delivery. The evaluation method was pre-registered before observing treatment data. Removing the threshold increased total spending, with growth occurring across all price points—not just low-value convenience items. Delivery efficiency improved as routes became denser. The net financial impact was neutral: incremental margins offset higher fulfillment costs. These results suggest that purchase thresholds, while improving unit economics, may suppress demand. Friction removal can be self-financing through density-driven economies of scale, with implications for subsame day delivery infrastructure expansion.

Mentoring Matters: Ambassador Quality and New Hire Productivity in Fulfillment Centers: Working Paper
We examine the relationship between ambassador characteristics and new hire performance outcomes in Amazon fulfillment centers, drawing on established education literature on teacher value added. Using data from Generation Type 9 AR Sortable warehouses across North America, we develop a framework for predicting and measuring ambassador quality, demonstrate the effect of ambassador quality on new hire productivity and retention, and propose a model for optimizing ambassador allocation. We find that higher quality ambassadors improve new hire productivity and reduce attrition and injuries. These effects translate to substantial savings through both increased productivity and reduced turnover costs.